Lecture Notes IV

Course: Analytical Mechanics

Topic: Central forces and the Kepler problem

We will study the central force problem and in particular, Kepler's problem, using the Lagrangian formalism and then making the contrast with the Newtonian formalism.

Central Forces

Consider a particle of mass m in space. Suppose that the applied force on m is a function of the distance to a given fixed point only, if such a point exists, the force is called central and the point is called the centre of the force (see figure below)

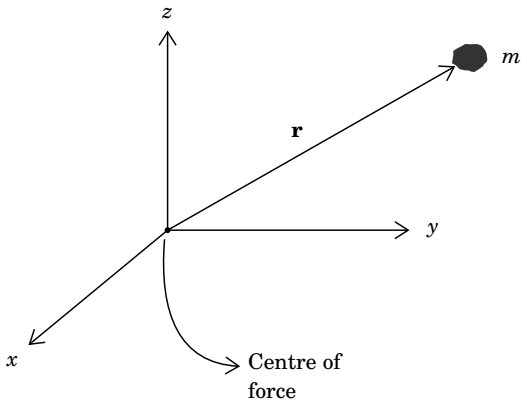


Figure 1: A particle of mass m located at position $\mathbf{r}$ with respect to the centre of force, placed at the origin of the Cartesian axes.

This way, the applied force on m can be written as

$$\mathbf{F}^{(a)}(r) = f(r)\hat{\mathbf{r}}, \quad r = ||\mathbf{r}||, \quad \hat{\mathbf{r}} = \frac{\mathbf{r}}{r}$$

and using the definition of a generalised force, and a given minimal set (q_1, q_2, q_3) of generalised coordinates, one can show that

$$Q_i = \mathbf{F}^{(a)} \cdot \frac{\partial \mathbf{r}}{\partial q_i} = f(r)\frac{\partial r}{\partial q_i}$$

so that for any f continuous in r we define $V(r) = -\int_a^r f$ and hence $f(r) = -\mathrm{d}V(r)/\mathrm{d}r$, this way

$$\frac{\partial V(r)}{\partial q_i} = -\frac{\mathrm{d}V(r)}{\mathrm{d}r}\frac{\partial r}{\partial q_i} = f(r)\frac{\partial r}{\partial q_i} = Q_i.$$

This shows that the generalised forces are derivable from a potential, so the Lagrangian has the standard form $L = T - V$, which on using spherical coordinates we get

$$L = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\dot{\phi}^2) - V(r)$$

so clearly, p_ϕ (generalised momentum conjugate to ϕ)

$$p_\phi = mr^2\sin^2\theta\dot{\phi} \equiv N$$

is a conserved quantity. The Euler-Lagrange equations on θ are thus

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(mr^2\dot{\theta}\right) = mr^2\sin\theta\cos\theta\dot{\phi}^2 = mr^2\sin\theta\cos\theta\frac{N^2}{m^2r^4\sin^4\theta}, \tag{1}$$

and since the coordinates (r,θ,ϕ) are no longer independent, and p_ϕ is the conserved quantity that connect the three quantities, the degrees of freedom are reduced to 2, and thus we have that $r = r(\phi), \theta = \theta(\phi)$ so that

$$\frac{\mathrm{d}\theta}{\mathrm{d}t} = \frac{\mathrm{d}\theta}{\mathrm{d}\phi}\frac{\mathrm{d}\phi}{\mathrm{d}t} = \frac{N}{mr^2\sin^2\theta}\frac{\mathrm{d}\theta}{\mathrm{d}\phi}$$

and therefore (1) becomes

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{N}{\sin^2\theta}\frac{\mathrm{d}\theta}{\mathrm{d}\phi}\right) = \frac{\cos\theta N^2}{mr^2\sin^3\theta},$$

but the term in the parentheses on the LHS is a function of θ only so we can write

$$\frac{N}{mr^2\sin^2\theta}\frac{\mathrm{d}}{\mathrm{d}\phi}\left(\frac{N}{\sin^2\theta}\frac{\mathrm{d}\theta}{\mathrm{d}\phi}\right) = \cos\theta\frac{N^2}{mr^2\sin^3\theta},$$

or, equivalently,

$$\frac{\mathrm{d}}{\mathrm{d}\phi}\left(csc^2\theta\frac{\mathrm{d}\theta}{\mathrm{d}\phi}\right) = \cot\theta \iff -\frac{\mathrm{d}^2}{\mathrm{d}\phi^2}(\cot\theta) = \cot\theta,$$

where the last equivalency comes from the chain rule and the fact that $\mathrm{d}\cot\theta/\mathrm{d}\theta = -\csc^2\theta$. The previous equation is a differential equation whose solution is a linear combination of sines and cosines, so

$$\cot\theta = A\sin\phi + B\cos\phi$$

so that multiplying both sides by $r\sin\theta$ we get

$$r\cos\theta = Ar\sin\theta\sin\phi + Br\sin\theta\cos\phi$$

and, equivalently,

$$z = Ay + Bx, \tag{2}$$

which is the equation of a plane passing through the origin (in this case, the centre of the force) in space, meaning that the movement only takes place in a plane determined by the equation $z = Ay + Bx$. Using the Newtonian formalism we see that by defining the vector $\mathbf{L} = \mathbf{r}\times\mathbf{p}$, $\mathbf{p} = m\dot{\mathbf{r}}$ we see that

$$\frac{\mathrm{d}\mathbf{L}}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}(\mathbf{r}\times\mathbf{p}) = \dot{\mathbf{r}}\times\mathbf{p} + \mathbf{r}\times\dot{\mathbf{p}} = \mathbf{r}\times\dot{\mathbf{p}} = \mathbf{r}\times\mathbf{F} = \mathbf{r}\times f(r)\frac{\mathbf{r}}{r} = \frac{f(r)}{r}\mathbf{r}\times\mathbf{r} = \mathbf{0}$$

meaning that $\mathbf{L}$ is a constant vector, which by its definition is perpendicular to $\mathbf{r}$ and $\mathbf{p}$ at all times, so the movement takes place only in a plane defined by the equation $L_x x + L_y y + L_z z = 0$, this gives us directly more information than before, since comparing this to (2) we see that $A = -L_x/L_z, B = -L_y/L_z$, i.e., the constants A,B are related to the (conserved) components of the angular momentum, and in particular, the z component is $L_z = xp_y - yp_x = mx\dot{y} - my\dot{x} = mr^2\sin^2\theta\dot{\phi}$ (the conserved quantity from the Lagrangian treatment). For deducing such a fact, the Newtonian formalism is way superior to the analysis of the central force problem than the Lagrangian one. Using the previous results we can go back to the Lagrangian formalism and since the motion takes place only in a plane, we may use a new minimal set (ρ,ψ) of polar coordinates in the plane where the motion takes place (see figure below)
The Lagrangian will thus have the form

$$L = \frac{1}{2}m(\dot{\rho}^2 + \rho^2\dot{\psi}^2) - V(\rho)$$

and again, we get that p_ψ is a conserved quantity

$$p_\psi = m\rho^2\dot{\psi} \equiv \ell$$

and hence we will have $\rho = \rho(\psi)$, we also see that since $\partial L/\partial t = 0$, then the Jacobi's integral is a constant of the motion and given that L is quadratic in the velocities, J is the energy of the system,

$$E = \frac{1}{2}m(\dot{\rho}^2 + \rho^2\dot{\psi}^2) + V(\rho) = \frac{1}{2}m\dot{\rho}^2 + \frac{\ell^2}{2m\rho^2} + V(\rho).$$

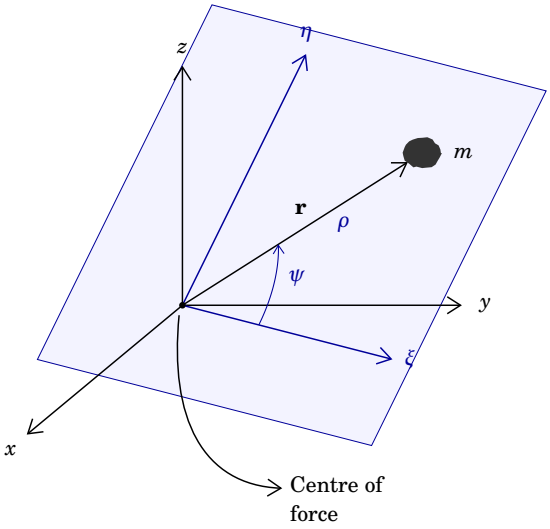


Figure 2: The motion takes place on a plane passing through the origin, with orthogonal axes ξ, η, on which the polar coordinates (ρ, ψ) are defined.

Given that the term $\ell^2/2m\rho^2$ (usually called the **centrifugal force** term) is a function of ρ only, we usually call $V(\rho) + \ell^2/2m\rho^2$ the *effective potential* and write $V_{eff}(\rho) = V(\rho) + \ell^2/2m\rho^2$. This way,

$$\frac{\mathrm{d}\rho}{\mathrm{d}t} = \pm\sqrt{\frac{2}{m}\big(E - V_{eff}(\rho)\big)}, \tag{3}$$

which on integrating yields

$$\pm\int \mathrm{d}t = \int \mathrm{d}\rho \frac{1}{\sqrt{\frac{2}{m}\big(E - V_{eff}(\rho)\big)}},$$

whence depending on the form of a given V, one may analytically find an expression $t(\rho)$ and luckily, invert the equation to get $\rho(t)$, which is the desired function. Equation (3) can be rewritten in terms of ψ by using the fact that

$$\frac{\mathrm{d}\rho}{\mathrm{d}t} = \frac{\mathrm{d}\rho}{\mathrm{d}\psi}\frac{\mathrm{d}\psi}{\mathrm{d}t} = \frac{\ell}{m\rho^2}\frac{\mathrm{d}\rho}{\mathrm{d}\psi},$$

to get

$$\frac{\ell}{m\rho^2}\frac{\mathrm{d}\rho}{\mathrm{d}\psi} = \pm\sqrt{\frac{2}{m}\big(E - V_{eff}(\rho)\big)}$$

or, integrating,

$$\int \mathrm{d}\rho \frac{\ell}{m\rho^2\sqrt{\frac{2}{m}\left(E - V(\rho) - \frac{\ell^2}{2m\rho^2}\right)}} = \pm\int \mathrm{d}\psi. \tag{4}$$

Kepler's Problem

Using the results of the previous section, we will now study the particular choice of $V(r) = -k/r$, sometimes called **Kepler's potential**, using (4) and the fact that $r = \rho$ we get

$$\int \mathrm{d}\rho \frac{\ell}{m\rho^2\sqrt{\frac{2}{m}\left(E + \frac{k}{\rho} - \frac{\ell^2}{2m\rho^2}\right)}} = \int \mathrm{d}\rho \frac{\ell}{\rho^2\sqrt{2mE + \frac{2mk}{\rho} - \frac{\ell^2}{\rho^2}}} = \pm\int \mathrm{d}\psi,$$

which clearly suggests the change of variable $\varrho = 1/\rho$, yielding

$$-\int \mathrm{d}\varrho \frac{\ell}{\sqrt{2mE + 2mk\varrho - \ell^2\varrho^2}} = -\int \mathrm{d}\varrho \frac{1}{\sqrt{\frac{2mE}{\ell^2} + \frac{2mk\varrho}{\ell^2} - \varrho^2}},$$

and completing the square in the denominator we get

$$-\int \mathrm{d}\varrho \frac{1}{\sqrt{\frac{2mE}{\ell^2}-\left(\varrho-\frac{mk}{\ell^2}\right)^2+\frac{m^2k^2}{\ell^4}}}$$

so defining $a^2 = 2mE/\ell^2 + m^2k^2/\ell^4$ suggests the new change of variable $a\cos\Omega = \varrho - mk/\ell^2$, this way the integral becomes

$$\int \mathrm{d}\Omega = \Omega = \pm\int \mathrm{d}\psi$$

and therefore $\Omega = \pm(\psi - \psi_0)$ where ψ_0 is a constant of integration, we now notice that

$$\cos\Omega = \cos\big(\pm(\psi-\psi_0)\big) = \cos\big(\psi-\psi_0\big),$$

thus going back to the original variables ρ, ψ

$$\frac{\varrho - mk/\ell^2}{a} = \cos\big(\psi-\psi_0\big),$$

whence

$$\frac{1}{\rho} = a\cos\big(\psi-\psi_0\big)+\frac{mk}{\ell^2} = \sqrt{\frac{2mE}{\ell^2}+\frac{m^2k^2}{\ell^4}}\cos\big(\psi-\psi_0\big)+\frac{mk}{\ell^2} = \frac{mk}{\ell^2}\sqrt{\frac{2mE\ell^4}{m^2k^2\ell^2}+1}\ \cos\big(\psi-\psi_0\big)+\frac{mk}{\ell^2},$$

and defining $e = \sqrt{\dfrac{2E\ell^2}{mk^2}+1}$ we get

$$\frac{1}{\rho} = \frac{mk}{\ell^2}\big(e\cos\big(\psi-\psi_0\big)+1\big). \tag{5}$$

Recall that $\xi = \rho\cos\psi$ and $\eta = \rho\sin\psi$ (see Figure 2) so that redefining $\psi \to \psi - \psi_0$ (now ψ is measured from an initial angle ψ_0 and not from ξ) we get that (5) is just

$$\frac{\ell^2}{mk\sqrt{\xi^2+\eta^2}} = e\frac{\xi}{\sqrt{\xi^2+\eta^2}}+1,$$

and multiplying both sides by $\sqrt{\xi^2+\eta^2}$ we get

$$\frac{\ell^2}{mk} = e\xi+\sqrt{\xi^2+\eta^2} \iff \sqrt{\xi^2+\eta^2} = \frac{\ell^2}{mk}-e\xi,$$

and squaring both sides

$$\xi^2+\eta^2 = \frac{\ell^4}{m^2k^2}+e^2\xi^2-\frac{2e\ell^2\xi}{mk}, \tag{$*$}$$

which is equivalent to

$$\xi^2(1-e^2)+\frac{2e\ell^2\xi}{mk}+\eta^2 = (1-e^2)\left(\xi^2+\frac{2e\ell^2\xi}{mk(1-e^2)}\right)+\eta^2 = \frac{\ell^4}{m^2k^2},$$

and completing the square on ξ we have

$$(1-e^2)\left(\left[\xi+\frac{e\ell^2}{mk(1-e^2)}\right]^2-\frac{e^2\ell^4}{m^2k^2(1-e^2)^2}\right)+\eta^2 = \frac{\ell^4}{m^2k^2},$$

so that

$$(1-e^2)\left(\xi+\frac{e\ell^2}{mk(1-e^2)}\right)+\eta^2 = \frac{\ell^4}{m^2k^2}+\frac{e^2\ell^4}{m^2k^2(1-e^2)} = \frac{\ell^4}{m^2k^2}\left\{1+\frac{e^2}{(1-e^2)}\right\} = \frac{\ell^4}{m^2k^2(1-e^2)}$$

and dividing both sides by the constant factor,

$$\frac{(1-e^2)^2m^2k^2}{\ell^4}\left(\xi+\frac{e\ell^2}{mk(1-e^2)}\right)+\frac{m^2k^2(1-e^2)}{\ell^4}\eta^2 = 1. \tag{6}$$

We now split the treatment in different cases

i) For the case $e < 1$, $1 - e^2 > 0$ and thus we can define

$$\alpha^2 = \frac{\ell^4}{(1-e^2)^2 m^2 k^2}, \quad \beta^2 = \frac{\ell^4}{m^2 k^2 (1-e^2)}, \quad h = -\frac{e\ell^2}{mk(1-e^2)}$$

and thus write (6) as

$$\frac{(\xi - h)^2}{\alpha^2} + \frac{\eta^2}{\beta^2} = 1,$$

which shows that in this case we have an ellipse in the $\xi\eta$-plane.

ii) If $e > 1$ then $1 - e^2 < 0$, so we introduce a minus sign in the definition of β^2 (it would be complex otherwise) to get

$$\frac{(\xi - h)^2}{\alpha^2} - \frac{\eta^2}{\beta^2} = 1,$$

which shows that in this case we have a hyperbola in the $\xi\eta$-plane.

iii) if $e = 1$, we have to go back all the way to $(*)$ and notice that instead of factoring out $1 - e^2$, we can simply write

$$\eta^2 = \frac{\ell^4}{m^2 k^2} - \frac{2\ell^2 \xi}{mk}$$

which shows that in this case we have a parabola in the $\xi\eta$-plane.

It is left as an exercise to find the value of e (and thus of E) for which the motion that takes place in the $\xi\eta$-plane is a circle, and then find the radius of such a circle.